Technical Reproducibility Validation Report

GangSTR

Version v2.5
Autosomal STR

Verification date	2026-11-06
Repository commit	6ea9b2b8daca51dcab1f0e46210622b94b52ff17
Result	**Not verified as documented**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/gangstr
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Not verified as documented**
REACHED	Runs + Plausible output, 6/6 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. What STRhub had to do to run this tool

STRhub wrote its own recipe for this tool, an environment and a command of its own rather than the repository's instructions, and ran that. Each item below is something a first-time user following the README would have to work out for themselves, and so a recommendation to the author. This run does not change the tool's label: the label is what happens as it is in the repository.

– Installs GangSTR 2.5.0 from bioconda, a prebuilt binary. Instead of: cmake .. && make from the source at the pinned commit, as the README documents. The documented build stops in htslib: the repository's CMakeLists runs autoreconf without -i (config.sub/config.guess missing). What ran is bioconda's build of 2.5.0, not this commit's source.

– Supplies the regions BED from STRhub's library and the command-line flags STRhub chose for its reference slice. Instead of: A regions file and a command the README leaves to the user.

With those changes, the run reached: **Runs + Plausible output**.

3. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Starts	The installed program answers --help in the built environment	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	gangstr.log-external.txt	

4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · GangSTR v2.5 · commit 6ea9b2b
$ GangSTR \
--bam /data/in/input.bam \
--ref /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--out /data/out/output
```

5. Reproducibility Metadata

Tool	GangSTR
Version	v2.5
Repository	https://github.com/gymreklab/gangstr
Commit (pinned)	6ea9b2b8daca51dcab1f0e46210622b94b52ff17
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-11-06 16:37:44 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42253585030

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

6. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

7. Output Content Evidence

Output file	output.vcf
Format	VCF
Sequence records	24
Distinct STR loci	24
Total reads	10,738
Max depth (single locus)	757

8. Verification Data

The repository ships example data (10 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · SE33 · TH01 · TPOX · vWA
Regions BED	Provided by STRhub

9. Verification Matrix

PASS	Run on repository test data	—
	Run on reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– A container environment and a command written by STRhub, not taken from the repository's instructions; what they do differently is listed under "What STRhub had to do to run this tool".

11. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

12. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

13. Conclusion

◆ **Runs end-to-end**

GangSTR v2.5 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

◆ **Produces plausible output**

The tool generated a structurally valid VCF output file with genotype calls across 24 target forensic STR loci, with a maximum read depth of 757 at D18S51.

◆ **Verdict: Runs**

The tool installed and its run produced its documented output.

◆ **Repository test data not used**

The repository ships example data (10 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
D18S51	757		**D13S317**	442	
D16S539	618		**CSF1PO**	435	
PentaD	613		**D22S1045**	429	
D5S818	589		**D19S433**	361	
D6S1043	585		**TPOX**	328	
D7S820	579		**SE33**	328	
D3S1358	575		**TH01**	306	
D8S1179	558		**D21S11**	282	
D2S441	517		**D12S391**	281	
vWA	495		**D1S1656**	268	
D10S1248	460		**FGA**	254	
PentaE	456		**D2S1338**	222	